

Saksham Sinha

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Education

SRM Institute of Science and Technology

B.Tech in Electronics and Communication Engineering

Sri Sri Suryadeo Singh Smriti Gurukulam (CBSE)

Dhanbad Public School (CBSE)

Expected 2027

CGPA: 9.56/10

Class XII: 78.4%

Class X: 92%

Experience

Backend Developer Intern

June 2026 – August 2026

Little Story Live-Gen Studio Pvt. Ltd.

Remote/Hybrid

- Developed RESTful APIs (FastAPI, PostgreSQL) for a live-gen AI media platform, building robust endpoints for media asset uploads, metadata storage, and end-to-end publishing workflows.
- Implemented secure JWT authentication and **Role-Based Access Control (Admin / Creator / User)**, ensuring appropriate data isolation across different stakeholder tiers.
- Offloaded heavy AI media generation tasks by integrating asynchronous Redis task queues, reducing average API latency by **25%**; containerized backend services with Docker for reliable deployment.

Technical Skills

Languages: C++, Go, Python, JavaScript, TypeScript, SQL

Frontend: React.js, Next.js, HTML, CSS, Tailwind CSS

Backend: FastAPI, Node.js, Express.js, REST APIs, WebSockets, Socket.io, Redis, PostgreSQL

Systems: Goroutines, POSIX threads, distributed coordination, PostgreSQL row-level locking

Tools: Docker, Git, GitHub, Linux, Postman, Mediasoup, WebRTC

Core CS: Data Structures & Algorithms, OOP, Operating Systems, DBMS, Computer Networks

Projects

Real-Time SFU Diagnostics Platform

GitHub

Node.js, Express, Mediasoup, Socket.io, Next.js, Docker

- Built a remote field-diagnostics platform routing WebRTC media through a custom **Mediasoup SFU** — cutting client bandwidth **60%** vs. P2P mesh across 50+ concurrent sessions, with live FPS / latency / jitter telemetry per session.
- Added an **AR diagnostic pointer**: agents click the live video feed; the platform calculates relative X/Y coordinates and injects a visual marker on the technician's screen over WebSockets — reducing connection drops **40%** under packet loss via dynamic bitrate adaptation.

Distributed Job Scheduler

GitHub

Go, JavaScript, Python, Docker, PostgreSQL

- Built a distributed job scheduler with a **Go worker pool** and FastAPI control plane; used PostgreSQL FOR UPDATE SKIP LOCKED for contention-free job dispatch — zero duplicate claims across concurrent workers in integration tests.
- Implemented **per-queue concurrency limits**, exponential retry/backoff, dead-letter queue, and worker heartbeats with automatic recovery from stale workers.

Satya – Hybrid Fake News Detector

GitHub

Python, FastAPI, TypeScript, Gemini API, TF-IDF, Logistic Regression, MLP

- Achieved **94% validation accuracy** using a hybrid NLP pipeline combining TF-IDF, Logistic Regression, and MLP Neural Networks.
- Deployed an asynchronous FastAPI backend and TypeScript frontend with Gemini API integration for real-time explainable inference with **sub-second latency**.

Research & Publications

Plant Disease Severity Segmentation using nnU-Net

GitHub

Python, PyTorch, nnU-Net v2, SegFormer, PlantSeg

First Author — Manuscript in Preparation for Journal Submission

- Engineered an **nnU-Net v2** semantic segmentation pipeline with channel-wise normalization, achieving **79.38% pixel accuracy** across 11,458 image-mask pairs.
- Benchmarked nnU-Net against **SegFormer MIT-B0** and **ResNet18** across Dice, IoU, Precision, and Recall to evaluate lesion localization.

Achievements

- Flipkart GRID 8.0** — Cleared 2 of 4 rounds
- LeetCode**: Global Rank 1,384, Weekly Contest 507